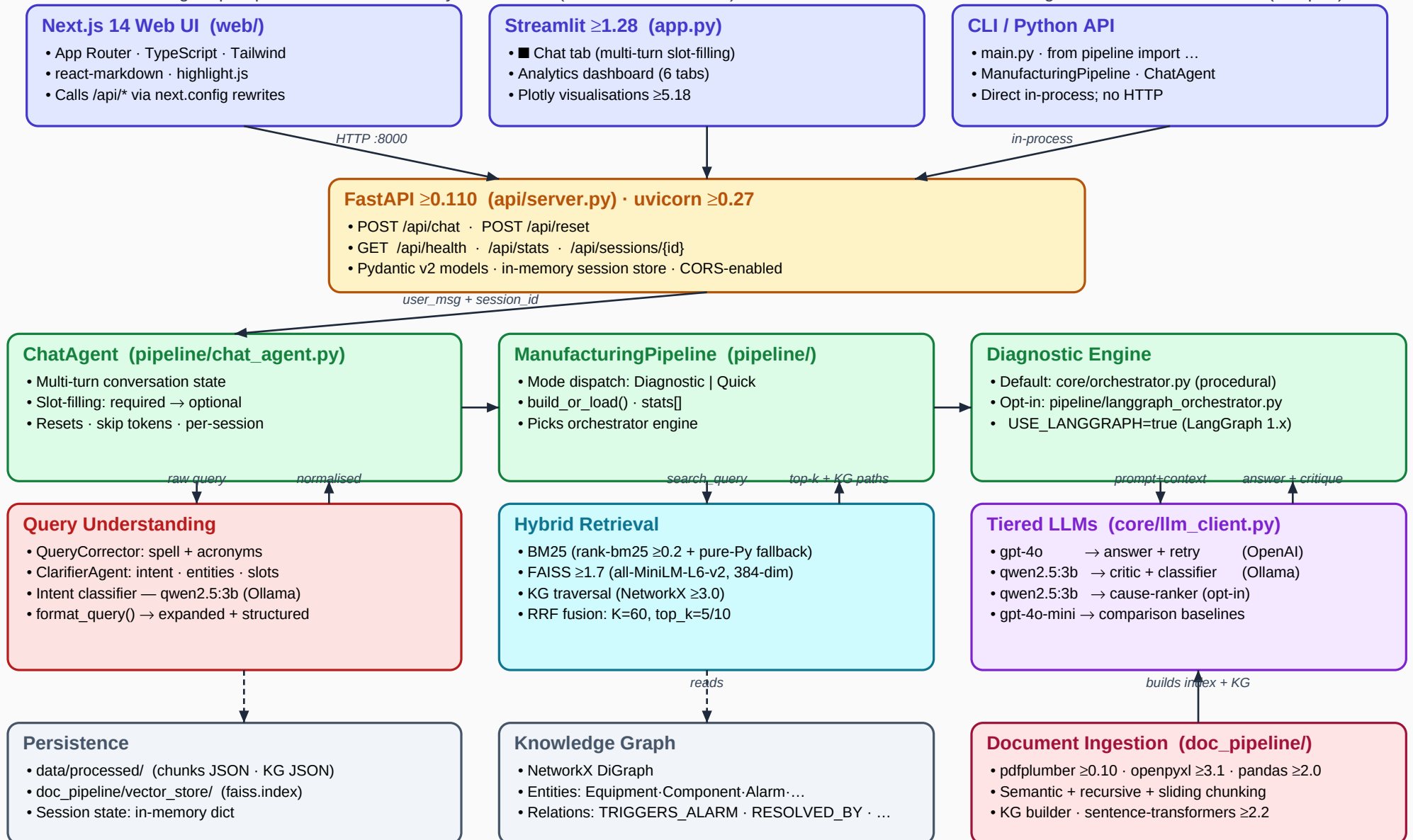


Hybrid GraphRAG Manufacturing — System Architecture

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Multi-turn chat • LangGraph-optional orchestration • Hybrid retrieval (BM25 + FAISS + KG) • Cause-ranker • Critic + deterministic guardrails • ERP/MES tools (see p. 8)



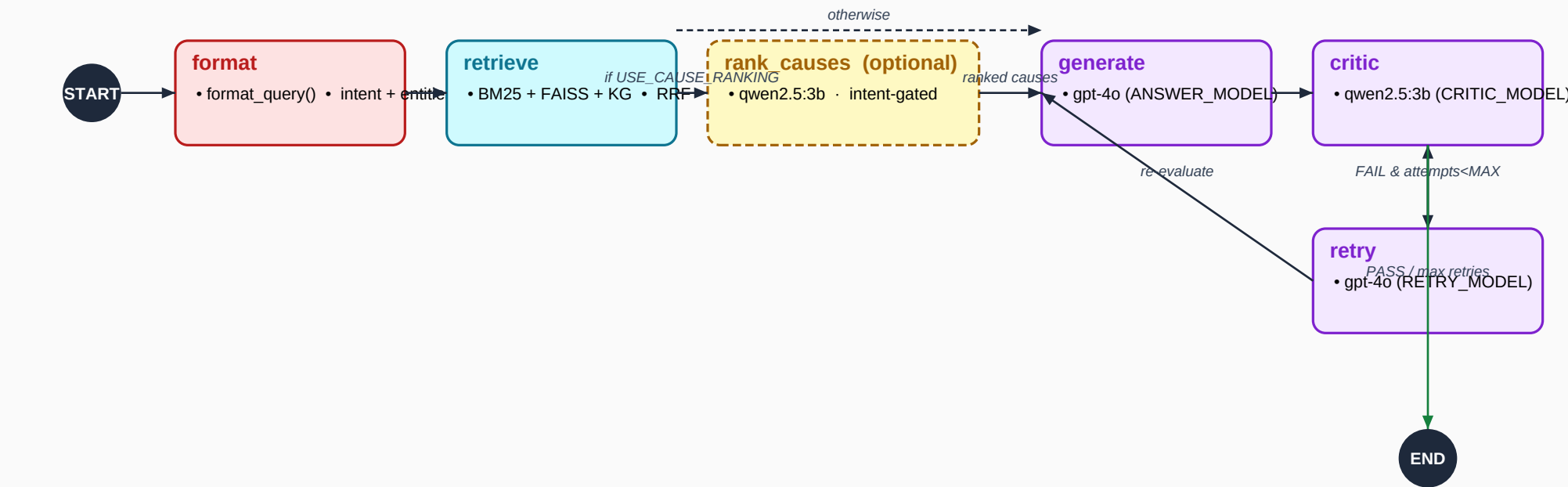
Legend

Solid arrow — runtime data flow

Dashed arrow — reads from / writes to persisted state

Diagnostic-Mode Flow — LangGraph Topology

USE_LANGGRAPH=true routes every diagnostic query through this StateGraph. The procedural orchestrator follows the same logical flow.



Node	Function	Model / Tool	Env vars	Latency p50	Cost p50
format	Intent + entity extraction; query expansion	regex → qwen2.5:3b fallback	CLASSIFY_MODEL	5–60 ms	~\$0 (local)
retrieve	BM25 + FAISS + KG traversal → RRF top-k	rank-bm25 + FAISS + NetworkX	TOP_K_RETRIEVAL=10 TOP_K_RERANK=5 RRF50–300 ms		\$0 (no LLM)
rank_causes (opt-...)	Score top-N root causes from evidence + KG	CAUSE_RANK_MODEL qwen2.5:3b	USE_CAUSE_RANKING CAUSE_RANK_TOP_K=5100–500 ms		\$0 (Ollama)
generate	First-pass evidence-grounded answer	ANSWER_MODEL gpt-4o	OPENAI_API_KEY	1.5–3 s	~\$0.0050
critic	Grounding / completeness / safety check	CRITIC_MODEL qwen2.5:3b	—	50–250 ms	\$0 (Ollama)
retry	Regenerate using critic feedback	RETRY_MODEL gpt-4o	MAX_CRITIC_RETRIES=2	1.5–3 s	~\$0.0050 (per r...

Cost & Latency Breakdown

Per-query estimates with the default tiered routing (answer = gpt-4o, critic = qwen2.5:3b on Ollama). Local models are free; OpenAI pricing per core/llm_client.py.

1. Per-mode summary (typical query, default config)

Mode	LLM calls per query	Tokens (in / out)	Cost / query	Latency p50	Notes
Quick Search	0	0 / 0	\$0.0000	120–400 ms	FAISS + embeddings only; w...
Diagnostic (default)	1 answer + 1 critic	~1500 / ~400 + ~700 / ~150	~\$0.0080	2.0–4.5 s	ANSWER on gpt-4o, CRITIC o...
Diagnostic + cause-ranker	+1 cause-ranker	+~1200 / ~250	+ \$0.0000	+100–500 ms	USE_CAUSE_RANKING=true; in...
Diagnostic worst-case	1 answer + 2 critics + 1 ret...	~3700 / ~950	~\$0.0160	4.5–9 s	Critic FAILs once, retry r...
Chat (multi-turn)	Same as Diagnostic	Same as Diagnostic	Same as Diagnos...	Same as Diagnost...	Plus per-turn slot-filling...
Classical RAG (baseline)	1	~1500 / ~400	~\$0.0005	1.0–2.0 s	FAISS-only retrieval, gpt-...
Direct LLM (baseline)	1	~150 / ~400	~\$0.0003	0.8–1.5 s	No retrieval; gpt-4o-mini ...

2. Per-stage detail for Diagnostic mode (USE_CAUSE_RANKING=true, 1 retry path)

Stage	Model	Provider	Tokens (in / out)	Cost / call	Latency p50	Mandatory?
format	regex → qwen2.5:3b	Ollama (fallback)	0–250 / 0–80	\$0	5–60 ms	yes
retrieve	—	FAISS + BM25 + KG	0 / 0	\$0	50–300 ms	yes
rank_causes	qwen2.5:3b	Ollama	~1200 / ~250	\$0	100–500 ms	USE_CAUSE_RANKIN...
generate	gpt-4o	OpenAI	~1500 / ~400	~\$0.0050	1.5–3 s	yes
critic (1st)	qwen2.5:3b	Ollama	~700 / ~150	\$0	50–250 ms	yes
retry (if FAIL)	gpt-4o	OpenAI	~1700 / ~400	~\$0.0050	1.5–3 s	FAIL & attempts<...
critic (2nd)	qwen2.5:3b	Ollama	~700 / ~150	\$0	50–250 ms	after retry

3. Cloud vs local pricing (per 1k tokens, from core/llm_client.py)

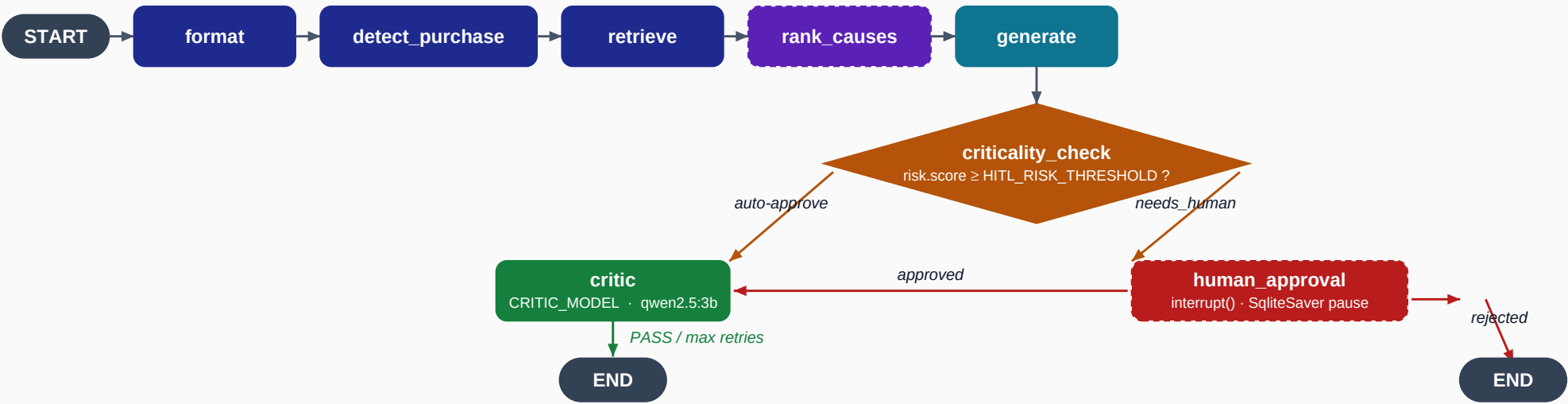
Model role	Default	OpenAI in / out	Local equivalent	Switch via .env	Net effect
Answer / Retry	gpt-4o	\$0.0025 / \$0.010	qwen2.5:7b on Ollama	ANSWER_MODEL=qwen2.5:7b	100% free, slower late...
Critic	qwen2.5:3b	n/a (local)	—	CRITIC_MODEL=gpt-4o-mini	\$0 → ~\$0.0001 / call
Cause-ranker (opt-in)	qwen2.5:3b	n/a (local)	—	CAUSE_RANK_MODEL=gpt-4o-mini	\$0 → ~\$0.0002 / call
Comparison (Direct/Classical)	gpt-4o-mini	\$0.00015 / \$0.0006	qwen2.5:3b on Ollama	DIRECT_LLM_MODEL / CLASSICAL_RAG_MODEL	Free at slight quality...
Embeddings	MiniLM-L6-v2	n/a (local)	—	EMBEDDING_MODEL=BAAl/bge-small-en-v1.5	Rebuild FAISS after ch...

Human-in-the-Loop (HITL) Approval Gate

criticality_check + human_approval (interrupt) • SQLite checkpointer • audit log • USE_HITL=true

LangGraph topology with the HITL gate

START → linear pipeline → criticality_check decision → auto-approve OR human_approval (interrupt) → critic → END



Legend ■ deterministic ■ optional cause-ranker ■ answer LLM ■ decision (risk gate) ■ human interrupt ■ critic ■ terminal

Risk score drivers — core/criticality_classifier.py

Driver	Trigger	Score bump	Domain
safety_keyword:*	Substring of HITL_HIGH_RISK_KEYWORDS in query or proposed answer	0.55 + 0.05 / extra hit	diagnostic
high_risk_intent	Clarifier intent ∈ {shutdown, emergency, lockout_tagout, permit_to_work}	0.90	diagnostic
low_critic_confidence	Critic verdict.confidence < 0.5	0.30	diagnostic
purchase_value>threshold	PurchaseRequest.total_usd ≥ HITL_AUTO_APPROVE_BELOW_USD	0.70	purchase_request
single_source_vendor	KG flags the part with single_source=true	0.65	purchase_request
long_lead_time	lead_time_days > 7	0.55	purchase_request
class_A_equipment	Used by an Equipment node tagged criticality=A	0.70	purchase_request
llm_grader:*	Tier-2 LLM grader (only fires for inconclusive 0.3–0.7 band)	max(score, llm_score)	any

REST surface (api/server.py)

GET /api/approvals/pending	list paused HITL workflows
GET /api/approvals/{thread_id}	snapshot of one paused workflow
POST /api/approvals/{thread_id}/resume	{approved, approver, comments, edited_answer?}
GET /api/approvals/my	buckets pending / pending_for_me / actioned
GET /api/audit?limit=N&offset=M	recent decisions + approval-rate stats

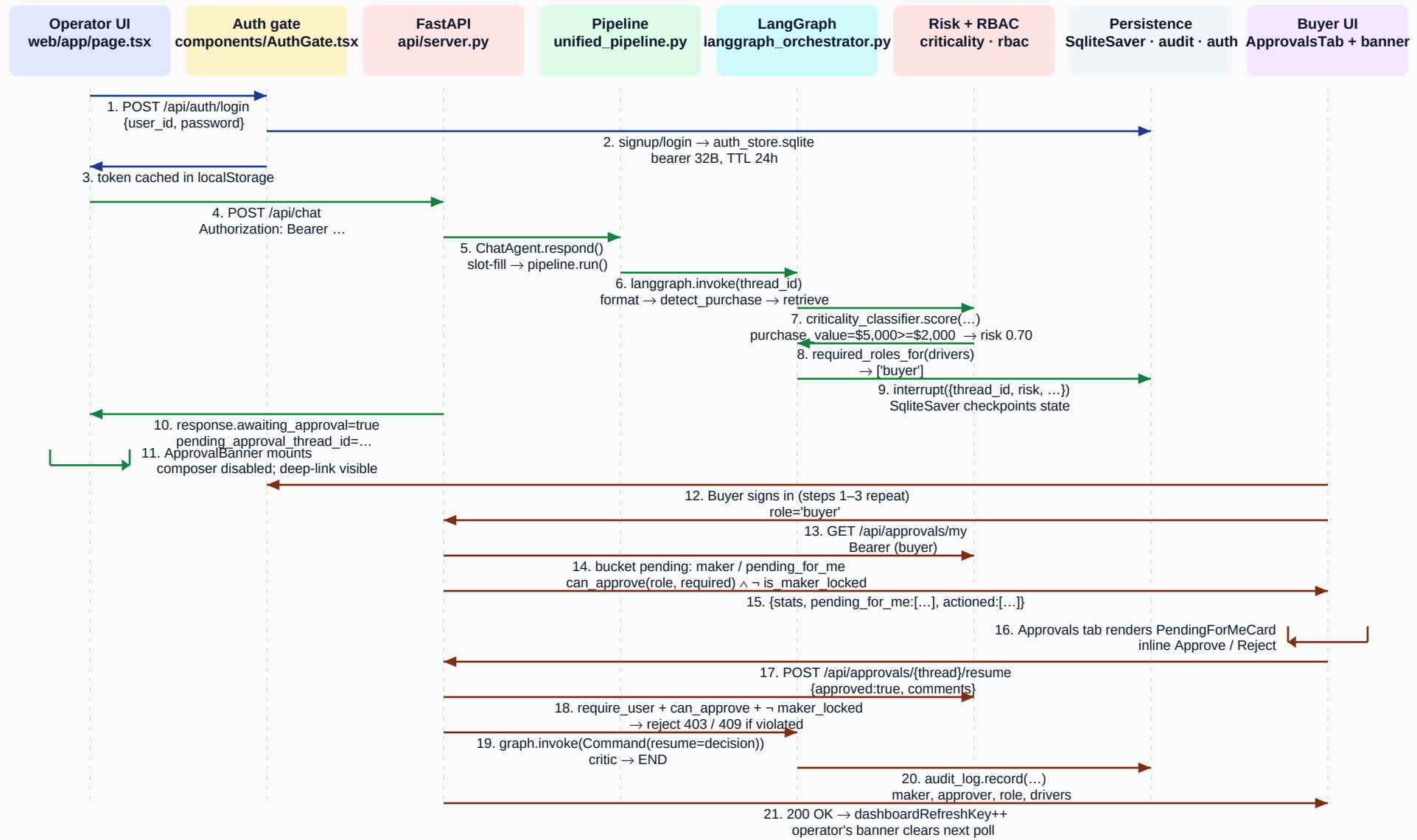
Pipeline status state machine

in_progress → awaiting_approval → complete (approved)
in_progress → awaiting_approval → rejected (approved=false)
in_progress → complete (auto-approve / USE_HITL=false)

Checkpoint: SqliteSaver (HITL_DB_PATH) — survives restarts.
Audit log: core/audit_log.py — append-only, one row per decision.

Low-Level Component Sequence — \$5,000 PO end-to-end

Operator submits a high-risk PO → LangGraph pauses at the HITL interrupt → buyer signs off from the Approvals tab. Each step names the file and wire payload.



- Phases**
- 1–3 Auth / token mint (24h TTL, stdlib PBKDF2-SHA256)
 - 4–11 Chat → risk → interrupt; state durable via SqliteSaver
 - 12–21 Buyer signs in, approves, audit row written, banner clears

Component Interaction Contracts

Every cross-component edge in the system. Use this page as the single source of truth when changing a payload, an auth check, or a failure mode.

1. UI → API (HTTP boundary)

From	To	Route / Payload	Auth	Fails when	p50
Next.js page.tsx	api/auth	POST /api/auth/login {user_id, password}	—	wrong password → 401	20–50 ms
Next.js page.tsx	api/server	POST /api/chat {session_id, message}	Bearer (opt)	422 / 500 on pipeline error	0.4–9 s
ApprovalsTab	api/server	GET /api/approvals/my	Bearer	401 on missing token	30–80 ms
PendingForMcCard	api/server	POST /api/approvals/{thread}/resume	Bearer (checker)	403 wrong role · 409 maker-lock · 404 un...	0.6–3.5 s
ApprovalBanner	api/server	GET /api/approvals/{thread_id}	Bearer (opt)	404 unknown thread	20–50 ms
Streamlit app.py	api/server	same endpoints (in-tab login)	Bearer (shared)	same as above	same

2. FastAPI → domain (in-process Python)

From	To	Call / contract	Notes	Side effects	p50
api/server	ChatAgent.respond	respond(session_id, msg, user) → ChatTurn	Slot fill → pipeline.run	stamps maker_user_id	0.4–9 s
ChatAgent	ManufacturingPipeline.run	run(query, mode='diagnostic') → Result	Intent + USE_LANGGRAPH	may pause at interrupt	0.4–9 s
ManufacturingPipeline	LangGraphOrchestrator	invoke(state, config={thread_id})	USE_LANGGRAPH=true	checkpoint per node	0.4–9 s
LangGraph node	criticality_classifier	score(query, answer, purchase_request) → Risk	Rules → drivers → score	—	1–5 ms
criticality_classifier	rbac.required_roles_for	required_roles_for(drivers, pr) → [role_id]	OR-set; stable order	—	<1 ms
api/server (resume)	rbac.can_approve / is_maker_loc...	two boolean guards before resume	403 / 409 if violated	—	<1 ms
LangGraph (resume)	audit_log.record	record(thread, decision, maker, approver, ro...	Append-only	SQLite write	1–4 ms

3. Domain → persistence

Component	Backend	Lifecycle	Failure mode	Drop-in replacement
auth_store	auth.sqlite (stdlib sqlite3)	open per call; 24h token TTL	delete file → re-seed accounts	OIDC · Argon2
SqliteSaver	audit.sqlite (lg_checkpoint_sqlite)	one checkpoint per node	missing pkg → MemorySaver (WARN)	Postgres saver
audit_log	audit.sqlite (append-only)	writes on every approve/reject	disk full → 500	Kafka topic
FAISS index	doc_pipeline/vector_store/*	rebuilt only when stale	missing → rebuild on boot	Qdrant / Pinecone
Knowledge graph	data/processed/knowledge_graph.json	rebuilt when input docs change	missing → rebuild on boot	Neo4j / TigerGraph
Session memory	in-memory dict (FastAPI process)	wiped on restart	crash → session lost (threads survive)	Redis

4. Authorisation policy on /api/approvals/{thread}/resume

1. Bearer token resolves to a User (else 401 unauthorised).
2. user.role ∈ required_roles (else 403 forbidden — wrong role).
3. user.user_id ≠ maker_user_id (else 409 conflict — maker locked).
4. thread_id is still pending (else 404 not found / 409 if already resolved).

Outcome: graph resumes → audit row written → stats refresh → both UIs refetch on next poll.

Role-Based Knowledge-Base Access (Document ACLs)

Three-tier classification • ContextVar-scoped retriever filter • smoke-test proven • core/document_acl.py

Public
tier id: "public" • everyone (incl. operators)

- SOPs (CNC machining, conveyor PM)
- Alarm-response procedures
- Equipment manuals (pumps, presses)
- Public safety bulletins
- Production / quality plan summaries

Restricted
tier id: "restricted" • every checker role (not operator)

- Regulatory incident response playbook
- Internal RCA (Q4 spindle-bearing case)
- Litigation-sensitive draft findings
- Detailed work orders + RCAs
- Lockout / permit-to-work records

Confidential
tier id: "confidential" • plant_manager + procurement_manager only

- Q1 2026 financial review (EBITDA, capex)
- Project Meridian M&A diligence paper
- Strategic supplier pricing (SKF, Siemens, Sandvik)
- Leadership succession + retention plan
- MFN clauses, walk-away prices

Role → tier read-sets

Each row is a sign up role. A filled dot means the retrievers will return chunks at that tier for the signed-in user.

role (rbac.py)		public	restricted	confidential
operator	line / control-room operator	●	●	●
shift_supervisor	shift supervisor	●	●	●
maintenance_planner	maintenance planner	●	●	●
maintenance_engineer	maintenance engineer	●	●	●
ehs_officer	EHS officer	●	●	●
quality_engineer	quality engineer	●	●	●
buyer	buyer (POs ≤ \$10k)	●	●	●
procurement_manager	procurement manager	●	●	●
plant_manager	plant manager	●	●	●

Ingest-time tagging (one folder convention, zero front-matter)

doc_pipeline/document_ingestion.py + chunking.py + core.document_acl.classify_from_path

doc_pipeline/input_docs/

- sop_cnc_machining.txt
- quality_control_manual.pdf
- production_metrics_q1_2026.xlsx
- restricted/
 - regulatory_incident_response_plan.txt
 - internal_incident_rca_2025_q4.txt
- management/
 - q1_2026_financial_review.txt
 - acquisition_target_assessment.txt
 - strategic_supplier_pricing.txt
 - leadership_succession_2026.txt

PUBLICPUBLICPUBLIC

RESTRICTEDRESTRICTED

CONFIDENTIALCONFIDENTIALCONFIDENTIAL

Query-time filter (zero parameter plumbing)

ContextVar set by FastAPI per request → retrievers honour it deep in the stack

Bearer token
POST /api/chat

with user classifications(role)
core/document_acl.py

HybridRetriever.retrieve()
BM25 ■ vector ■ graph → RRF

filter_chunks(fused)
drops chunks whose .classification ∉ read-set

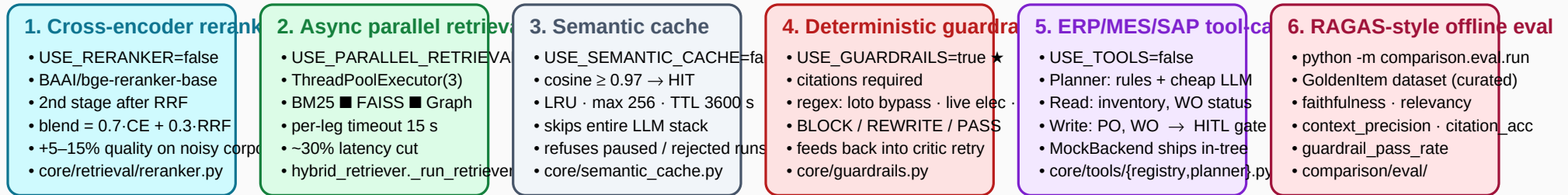
LLM prompt
evidence the user is entitled to see

Smoke-test proof — scripts/smoke_test_acl.py
Query: "Q1 2026 EBITDA and capex execution plan" → same FAISS index, two roles:
operator (public-only) → 0 confidential chunks; top-3: quality_control_manual.pdf, production_planning_report.pdf ×2
plant_manager (all tiers) → 5 confidential chunks; top-3: q1_2026_financial_review.txt ×3 (Header, Revenue & Margin, Capex Execution)

Advanced Patterns — rerank · cache · parallel · guardrails · tools · offline eval

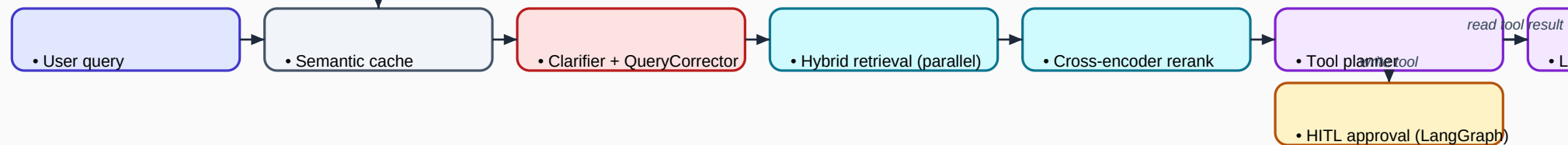
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Six production-hardening layers wrapping the core Hybrid GraphRAG engine. Each is gated by a single env flag.



★ safe-on by default on a fresh .env (run.sh enables parallel retrieval + guardrails; rerank · cache · tools stay opt-in to avoid surprise model downloads, memory pressure or write surface)

Updated request flow with all patterns engaged



Feature flags (config.py)

Flag	Default	Effect
<code>USE_RERANKER</code>	false	Cross-encoder rerank after RRF (5–15% quality lift)
<code>RERANKER_MODEL</code>	BAAI/bge-rerank...	Any HF cross-encoder; first run downloads it
<code>RERANK_CANDIDATE_POOL</code>	20	Pool forwarded to reranker; final cut uses <code>TOP_K_RERANK</code>
<code>RERANK_BLEND_WEIGHT</code>	0.7	$\text{Final} = w \cdot \text{CE} + (1-w) \cdot \text{RRF}$
<code>USE_PARALLEL_RETRIEVAL</code>	true ★	Fan BM25 / FAISS / Graph across a thread pool
<code>PARALLEL_RETRIEVAL_TIMEOUT_S</code>	15.0	Per-leg timeout (empty result on miss)
<code>USE_SEMANTIC_CACHE</code>	false	In-memory cosine cache; hit short-circuits LLM stack
<code>SEMANTIC_CACHE_THRESHOLD</code>	0.97	Cosine threshold for a hit
<code>SEMANTIC_CACHE_MAX_SIZE</code>	256	LRU eviction limit
<code>SEMANTIC_CACHE_TTL_SECONDS</code>	3600	Per-entry TTL
<code>USE_GUARDRAILS</code>	true ★	Citation + safety regex post-processor
<code>GUARDRAILS_REQUIRE_CITATIONS</code>	true	Reject answers without [source, chunk_id]
<code>GUARDRAILS_MIN_CITATIONS</code>	1	Minimum citations required
<code>GUARDRAILS_BLOCK_UNSAFE</code>	true	Hard-block LOTO bypass, live electrical, ...
<code>USE_TOOLS</code>	false	Enable ERP/MES tool planner + registry

RAGAS-style offline eval — `comparison/eval/` — one command

```
python -m comparison.eval.run --pipelines hybrid_graphrag classical_rag direct_llm --output comparison/eval/report.md --json-output comparison/eval/report.json
```

Metrics: faithfulness · answer_relevancy · context_precision · citation_accuracy · must_mention_coverage · guardrail_pass_rate · forbidden_violation_rate. Exit code is non-zero when --min-faithfulness floor is breached — CI